AP CALCULUS AB · EXAMINATION REPOSITORY

Unit 5
Analytical Applications of Differentiation

100-Question Master Examination

- Questions 1–90: Multiple Choice (Full Priority Ratings)
- Questions 91–100: Free Response with Graphs, Tables, and AP Scoring Rubrics
- Complete Diagnostic Answer Key and Step-by-Step Solutions at the End

OFFICIAL AP CED ALIGNED · MIXED LEVEL · RIGOROUSLY VERIFIED

Examination Structure

This paper contains 100 continuously numbered questions. All answers, worked solutions, and scoring rubrics appear at the end of the document.

⭐ Priority Rating System

★★★★★	**Very Important**	Core exam skills: Mean Value Theorem, Absolute Extrema (Candidates Test), Concavity & Inflection, Second Derivative Test, Optimization.
★★★	**Mandatory**	Essential skills: Monotonicity intervals, First Derivative Test, connecting f, f′, and f″ graphs, and Rolle's Theorem.
★★	**Important**	Foundational practice: Critical point definitions, standard quadratic extrema, basic theorem hypotheses.

- **Questions 1–90:** Multiple choice, four options, exactly one correct answer.
- **Questions 91–100:** Free response, 9 points each, with analytical graphs and tables.
- **Difficulty Mix:** Easy 5 · Moderate 68 · Hard 17.
- **Calculator Policy:** No calculator required.
- **Answer Section:** Begins after Question 100 — answer key, then full MCQ solutions, then FRQ solutions with rubrics.

Topic Coverage (Unit 5.1 – 5.12)

5.1 Using the Mean Value Theorem

5.2 Extreme Value Theorem & Global Extrema

5.3 Determining Intervals of Increase/Decrease

5.4 First Derivative Test for Relative Extrema

5.5 Candidates Test for Absolute Extrema

5.6 Determining Concavity of Functions

5.7 Using Second Derivative Test for Extrema

5.8 Sketching Graphs of Functions and Derivatives

5.9 Connecting f, f′, and f″ Behavior

5.10 Introduction to Optimization Problems

5.11 Solving Optimization Problems

5.12 Exploring Behaviors of Implicit Relations

> **Directions:** Attempt all 100 questions before consulting the answer section. For free response questions, identify the theorem or derivative test used and state all hypotheses.

QUESTION 01

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $f(x)=x^2-4x$ on $[0,4]$, the value c guaranteed by the Mean Value Theorem is

A 4

B 1

C 2

D 0

QUESTION 02

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Which Mean Value Theorem hypothesis fails for $f(x)=|x|$ on $[-1,1]$?

A Continuity on $[-1,1]$

B Differentiability on $(-1,1)$

C Existence of f' at the endpoints

D Equality of endpoint values

QUESTION 03

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $f(x)=\sqrt{x}$ on $[1,4]$, the MVT value c is

A 5/2

B 9/4

C 3/2

D 3

QUESTION 04

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $f(x)=x^3$ on $[0,2]$, the MVT guarantees a c satisfying $f'(c)=4$. The value is

A $1/\sqrt{3}$

B $4/3$

C $2/\sqrt{3}$

D $\sqrt{3}$

QUESTION 05

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $f(x)=1/x$ on $[1,2]$, the value c guaranteed by the MVT is

A $\sqrt{2}$

B $1/\sqrt{2}$

C $3/2$

D 2

QUESTION 06

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Rolle's Theorem applied to $f(x)=x^2-6x+8$ on $[2,4]$ guarantees $f'(c)=0$ at

A $c=2$

B $c=3$

C $c=4$

D no such c

QUESTION 07

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For f(x)=ln x on [1,e], the MVT value c is

A e−1

B e

C 1/(e−1)

D √e

QUESTION 08

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For f(x)=sin x on [0,π], a value guaranteed by the MVT is

A π

B π/2

C π/4

D 3π/4

QUESTION 09

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If f is continuous on [a,b], which statement is guaranteed by the Extreme Value Theorem?

A f has an absolute maximum and an absolute minimum on [a,b]

B f has exactly one critical number

C f is increasing on [a,b]

D f is differentiable on (a,b)

QUESTION 10

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The function $f(x)=x$ on $(0,1)$ has

A an absolute minimum but no maximum

B an absolute maximum and minimum

C an absolute maximum but no minimum

D neither an absolute maximum nor an absolute minimum

QUESTION 11

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Which condition is essential for the Extreme Value Theorem on an interval $[a,b]$?

A $f(a)=f(b)$

B f is continuous on the closed interval $[a,b]$

C $f'(x)>0$ everywhere

D f is a polynomial

QUESTION 12

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For $f(x)=x^3$ on $[-1,1]$, the MVT equation $f'(c)=(f(1)-f(-1))/2$ has

A one solution $c=1/\sqrt{3}$ only

B exactly one solution $c=0$

C no solution

D two solutions $c=\pm 1/\sqrt{3}$

QUESTION 13

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $f'(x)=x(x-2)$, on which interval is f decreasing?

A $(2,\infty)$

B $(0,2)$

C $(-\infty,0)$

D $(-\infty,0)\cup(2,\infty)$

QUESTION 14

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $f'(x)=3(x-1)^2$, which statement is correct?

A f has a local minimum at $x=1$

B f increases on all real numbers

C f decreases on all real numbers

D f has a local maximum at $x=1$

QUESTION 15

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $f'(x)=-(x+2)^2$, which statement is correct?

A f is strictly decreasing on the real line

B f is strictly increasing on the real line

C f has a local maximum at $x=-2$

D f has a local minimum at $x=-2$

QUESTION 16

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Suppose f' is positive on $(-\infty,-1)$, negative on $(-1,3)$, and positive on $(3,\infty)$. Then f has

A. a local minimum at -1 and maximum at 3

B. local maxima at both points

C. local minima at both points

D. a local maximum at -1 and minimum at 3

QUESTION 17

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For $f(x)=x^3$, what happens at $x=0$?

A. f is decreasing on both sides

B. A local minimum occurs

C. A local maximum occurs

D. $f'(0)=0$, but there is no local extremum

QUESTION 18

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $f'(x)<0$ for $x<2$ and $f'(x)>0$ for $x>2$ near $x=2$, then f has

A. an inflection point at 2

B. a local minimum at 2

C. no critical number at 2

D. a local maximum at 2

QUESTION 19

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Let $f'(x)=(x+1)(x-2)^2$. Which is true?

A. f has a local maximum at $x=2$

B. f has a local minimum at $x=-1$ and no extremum at $x=2$

C. f has a local maximum at $x=-1$ and a local minimum at $x=2$

D. f has local minima at both points

QUESTION 20

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Which statement defines a critical number c of f?

A. $f(c)=0$

B. c is in the domain of f and $f'(c)=0$ or $f'(c)$ is undefined

C. $f'(c)$ is undefined only

D. $f'(c)=0$ only

QUESTION 21

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If f has a local extremum at an interior point c and is differentiable at c, then

A. $f''(c)=0$

B. $f'(c)$ is undefined

C. $f'(c)=0$

D. $f(c)=0$

QUESTION 22

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

For a differentiable function, f′ changes from positive to negative at c. Then

A f has a local minimum at c

B f is constant near c

C f has an inflection point at c

D f has a local maximum at c

QUESTION 23

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

For a differentiable function, f′ changes from negative to positive at c. Then

A f has a local maximum at c

B f has a local minimum at c

C f is decreasing on both sides

D f has no critical number at c

QUESTION 24

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Which example shows that $f'(c)=0$ does not by itself guarantee a local extremum?

A $f(x)=x^2$ at $c=0$

B $f(x)=-x^2$ at $c=0$

C $f(x)=x^3$ at $c=0$

D $f(x)=|x|$ at $c=0$

QUESTION 25

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The function $f(x)=|x-1|$ has a critical number at $x=1$ because

A. f is discontinuous at 1
B. $f'(1)=0$
C. $f(1)=0$
D. f is defined at 1 but $f'(1)$ does not exist

QUESTION 26

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $f'(x)>0$ throughout an interval, then f is

A. increasing on that interval
B. constant on that interval
C. concave up on that interval
D. decreasing on that interval

QUESTION 27

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $f'(x)<0$ throughout (a,b), then f is

A. increasing on (a,b)
B. concave up on (a,b)
C. constant on (a,b)
D. decreasing on (a,b)

QUESTION 28

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $f'<0$ on $(-\infty,2)$, $f'>0$ on $(2,5)$, and $f'<0$ on $(5,\infty)$, then

A maximum at both 2 and 5

B minimum at 2 and maximum at 5

C maximum at 2 and minimum at 5

D minimum at both 2 and 5

QUESTION 29

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $f'(x)=-(x-3)^2$, then at $x=3$ the function

A continues decreasing through a stationary point

B has a local maximum

C has a local minimum

D changes from decreasing to increasing

QUESTION 30

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A point where f' is undefined can be a local extremum if

A f' is positive on both sides

B the function is defined there and its nearby values support an extremum

C f'' is zero there

D the function is also undefined there

QUESTION 31

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

To find absolute extrema of a continuous function on [a,b], the Candidates Test compares f-values at

A endpoints and interior critical numbers

B only points where $f'=0$

C only inflection points

D only the endpoints

QUESTION 32

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For $f(x)=x^3-3x$ on $[-2,2]$, the absolute maximum value is

A 2

B -2

C 4

D 0

QUESTION 33

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For $f(x)=x^2-2x$ on $[0,3]$, the absolute minimum value is

A 1

B 0

C 3

D -1

QUESTION 34

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $f(x)=x^3-3x^2$ on $[0,3]$, the absolute minimum value is

A 0
B 3
C -4
D -3

QUESTION 35

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $f(x)=|x|$ on $[-1,2]$, which pair gives its absolute minimum and maximum values?

A minimum -1, maximum 2
B minimum -1, maximum 1
C minimum 0, maximum 1
D minimum 0, maximum 2

QUESTION 36

MODERATE · MULTIPLE CHOICE ★★ IMPORTANT

For $f(x)=x^2$ on $[-2,1]$, the absolute maximum occurs at

A both $x=0$ and $x=1$
B $x=-2$
C $x=0$
D $x=1$

QUESTION 37

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A continuous function on $[-3,5]$ is guaranteed to

A. attain both an absolute maximum and an absolute minimum

B. be differentiable everywhere

C. have exactly one local maximum

D. have an interior critical number

QUESTION 38

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If f is strictly increasing and continuous on [a,b], its absolute minimum and maximum occur respectively at

A. b and a

B. both at a

C. an interior critical number

D. a and b

QUESTION 39

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $f(x)=-x^2+2x$ on $[0,3]$, the absolute maximum value is

A. 3

B. 0

C. 1

D. -3

QUESTION 40

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $f(x)=x^2-4x$ on $[0,5]$, the absolute maximum value is

A 0

B 4

C -4

D 5

QUESTION 41

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $f(x)=-x^2+4x$ on $[0,5]$, the absolute minimum value is

A -5

B 0

C 4

D 5

QUESTION 42

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The function $f(x)=x$ on the open interval $(0,1)$ has

A absolute maximum 1 only

B neither an absolute maximum nor an absolute minimum

C absolute minimum 0 only

D absolute maximum 1 and minimum 0

QUESTION 43

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A continuous function on a closed interval is guaranteed to have

A. a maximum at an endpoint

B. an absolute maximum and an absolute minimum, not necessarily unique

C. a differentiable point where $f'=0$

D. a unique absolute maximum

QUESTION 44

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For f(x)=x^(2/3) on [−1,1], the absolute minimum is

A. 0 at x=0

B. −1 at x=0

C. 1 at both endpoints

D. no minimum

QUESTION 45

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For $f(x)=1/(1+x^2)$ on [−2,1], the absolute minimum is

A. 1 at x=0

B. 1/2 at x=1

C. 1/5 at x=−2

D. 0

QUESTION 46

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

In the Candidates Test for an absolute extremum on [a,b], an interior point where f′ is undefined

A must be ignored

B must be checked if it belongs to the domain of f

C cannot be a local extremum

D is automatically an absolute maximum

QUESTION 47

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $f''(x)=6x$, where is the graph of f concave down?

A all real x

B $(-\infty,0)$

C nowhere

D $(0,\infty)$

QUESTION 48

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $f''(x)=12(x-1)^2$, what can be concluded about concavity at $x=1$?

A There is an inflection point at 1

B There is a local maximum at 1

C There is no change in concavity at 1

D There is a local minimum at 1

QUESTION 49

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $f''(x)=x^2-4$, the graph is concave up on

A $(-2,\infty)$

B $(-\infty,2)$

C $(-2,2)$

D $(-\infty,-2)\cup(2,\infty)$

QUESTION 50

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $f''(x)=(x-1)(x+2)$, at which x-values does f change concavity?

A $x=-1$ and $x=2$

B $x=-2$ only

C $x=1$ only

D $x=-2$ and $x=1$

QUESTION 51

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

At an interior critical number c, if $f'(c)=0$ and $f''(c)>0$, the Second Derivative Test gives

A an inflection point

B a local maximum

C a local minimum

D no conclusion

QUESTION 52

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

At an interior critical number c, if $f'(c)=0$ and $f''(c)<0$, then f has

A a local maximum

B no conclusion

C an inflection point

D a local minimum

QUESTION 53

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

For $f(x)=x^4$ at $x=0$, $f'(0)=0$ and $f''(0)=0$. What does the Second Derivative Test conclude?

A Local minimum

B The test is inconclusive; further analysis shows a local minimum

C Inflection point

D Local maximum

QUESTION 54

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For $f(x)=x^3$ at $x=0$, $f'(0)=0$ and $f''(0)=0$. The point $x=0$ is

A a local maximum

B a local minimum

C an absolute maximum

D neither a local maximum nor a local minimum

QUESTION 55

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A point c is an inflection point of f when

A. $f''(c)=0$

B. the concavity of f changes at c and the point lies on the graph

C. f has a local extremum at c

D. $f'(c)=0$

QUESTION 56

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $f''(x)>0$ throughout an interval, then f' is

A. zero

B. constant

C. decreasing

D. increasing

QUESTION 57

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $f''(x)<0$ throughout an interval, then f is

A. concave up and f' is increasing

B. constant

C. increasing and concave up

D. concave down and f' is decreasing

QUESTION 58

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

For $f(x)=x^3$, which statement about $x=0$ is correct?

A. f has a local maximum

B. f has neither an inflection point nor an extremum

C. f has an inflection point but no local extremum

D. f has a local minimum

QUESTION 59

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

For $f(x)=x^4$, $f''(0)=0$. Which statement is true?

A. f is decreasing on both sides

B. $x=0$ must be an inflection point

C. $x=0$ is not an inflection point because concavity is upward on both sides

D. f has a local maximum at 0

QUESTION 60

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Suppose f'' is negative on $(-\infty,-1)$, positive on $(-1,2)$, and negative on $(2,\infty)$. The inflection numbers are

A. 2 only

B. -1 and 2

C. 0 only

D. -1 only

QUESTION 61

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If f′ is increasing on an interval, f is

A concave down there

B decreasing there

C concave up there

D constant there

QUESTION 62

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If f′ is decreasing on an interval, f is

A concave down there

B necessarily decreasing there

C concave up there

D necessarily increasing there

QUESTION 63

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

An inflection point can occur where f″ is undefined if

A f is discontinuous there

B f′ is positive on both sides

C f is continuous and its concavity changes across the point

D f″ is zero there

QUESTION 64

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $f'(c)=0$ and $f''(c)=0$, which conclusion is always valid?

A. c is an inflection point

B. The Second Derivative Test is inconclusive

C. f has a local minimum

D. f has a local maximum

QUESTION 65

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $f'(x)>0$ and $f''(x)<0$ on an interval, f is

A. decreasing and concave down

B. increasing and concave down

C. decreasing and concave up

D. increasing and concave up

QUESTION 66

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If f' changes from positive to negative at c, the graph of f has

A. a vertical tangent necessarily

B. an inflection point necessarily

C. a local maximum at c

D. a local minimum at c

QUESTION 67

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If f′ changes from negative to positive at c, the graph of f has

A a local minimum at c

B a point where f″ must be negative

C a local maximum at c

D a vertical asymptote

QUESTION 68

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If f′(c)=0 but f′ has the same sign on both sides of c, then f

A must have an inflection point at c

B must have a local maximum at c

C must have a local minimum at c

D does not have a local extremum at c by the First Derivative Test

QUESTION 69

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If f′(x)<0 and f″(x)>0 throughout an interval, then f is

A decreasing and concave up

B increasing and concave up

C increasing and concave down

D decreasing and concave down

QUESTION 70

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A graph of f′ crosses the x-axis from above to below at x=4. Assuming f is differentiable, f has

A. an inflection point at 4
B. a local maximum at 4
C. no critical number at 4
D. a local minimum at 4

QUESTION 71

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If f′ is positive and increasing on an interval, then f is

A. increasing and concave down
B. increasing and concave up
C. decreasing and concave up
D. decreasing and concave down

QUESTION 72

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If f′ is negative and decreasing on an interval, then f is

A. increasing and concave up
B. decreasing and concave up
C. increasing and concave down
D. decreasing and concave down

QUESTION 73

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A rectangle has perimeter 40 units. Its maximum possible area is

A 100

B 80

C 400

D 90

QUESTION 74

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Two positive numbers have sum 30. Their greatest possible product is

A 210

B 225

C 200

D 240

QUESTION 75

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A farmer uses 80 m of fencing for three sides of a rectangular field beside a straight river. The greatest possible area is

A 600 m^2

B 1,600 m^2

C 1,000 m^2

D 800 m^2

QUESTION 76

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Squares of side x are cut from the corners of a 12-by-12 sheet to make an open box. The volume is maximized when x=

A 2

B 1

C 3

D 6

QUESTION 77

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A symmetric rectangle is inscribed under $y=9-x^2$ and above the x-axis. If x is its positive half-width, its maximum area is

A $9\sqrt{3}$

B $12\sqrt{3}$

C 9

D 18

QUESTION 78

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Among closed right circular cylinders of a fixed positive volume, the one with least surface area has

A height equal to half its radius

B height equal to three times its radius

C height equal to twice its radius

D height equal to its radius

QUESTION 79

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For positive x and y with xy=36, the minimum possible value of x+y is

A 12
B 18
C 36
D 6

QUESTION 80

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A rectangle has area 64 square units. Its minimum possible perimeter is

A 32
B 16
C 64
D 24

QUESTION 81

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For x>0, y>0 with x+2y=30, the greatest possible product xy occurs at

A x=30, y=0
B x=20, y=5
C x=15, y=7.5
D x=10, y=10

QUESTION 82

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A rectangle is inscribed in a circle of radius 5. Its greatest possible area is

A $25\sqrt{2}$

B 50

C 20

D 25

QUESTION 83

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The point on $y=x^2$ closest to $(0,3)$ has x-coordinate(s)

A 0 only

B $\pm\sqrt{(5/2)}$

C $\pm\sqrt{5}$

D $\sqrt{(5/2)}$ only

QUESTION 84

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A company's profit is $P(x)=-x^2+40x-100$ for $0\le x\le 50$. The production level giving maximum profit is

A 50

B 20

C 10

D 40

QUESTION 85

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

For positive x with $2x+y=12$, the maximum value of x^2y is

A 96

B 72

C 64

D 48

QUESTION 86

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A rectangular pen beside a river has one side along the river and an interior divider perpendicular to it. If 100 m of fencing is available for the three perpendicular fences and the opposite side, the greatest area is

A $2{,}000/3$ m^2

B $1{,}000$ m^2

C $2{,}500/3$ m^2

D $1{,}250/3$ m^2

QUESTION 87

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A product is sold at price $p=120-2x$ dollars when x units are sold. The output maximizing revenue is

A $x=40$

B $x=30$

C $x=60$

D $x=20$

QUESTION 88

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A continuous function on $[-2,4]$ has f' positive on $(-2,0)$, negative on $(0,3)$, and positive on $(3,4)$. Its absolute maximum must occur at

A. $x=-2$ only
B. $x=4$ only
C. $x=3$ only
D. $x=0$ or an endpoint, so values must be compared

QUESTION 89

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A rectangular enclosure of area 400 m^2 has one side along a wall and needs fencing for the other three sides. The minimum fencing length occurs when the perpendicular sides are

A. $20\sqrt{2}$ m
B. 10 m
C. 20 m
D. $10\sqrt{2}$ m

QUESTION 90

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

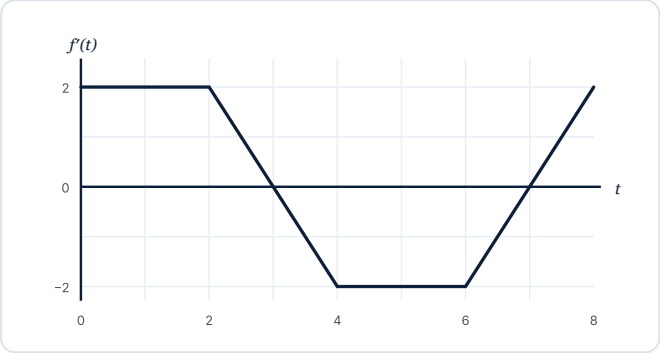
A rectangle has one vertex at the origin and opposite vertex (x,y) on $y=12-x$, with $x,y\geq 0$. The maximum area is

A. 30
B. 48
C. 36
D. 24

Question 91: Analyzing f′ from a Graph

★★★★★ VERY IMPORTANT 9 Points

The graph shows $f'(t)$ for $0 \le t \le 8$. The straight segments connect the plotted points. Given $f(0) = 0$.



Graph of f' on $0 \le t \le 8$; straight segments connect the plotted points.

(a) [2 points] State the intervals on which f is increasing and decreasing.

(b) [2 points] Find and classify the relative extrema of f in (0, 8).

(c) [3 points] Use signed areas under f′ to find f(8).

(d) [2 points] Find the absolute maximum and minimum values of f on [0, 8].

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

Question 92: Mean Value Theorem

★★★★★ VERY IMPORTANT | 9 Points

Let $f(t) = t^2 - 4t + 1$ on the closed interval $[0, 4]$.

(a) [2 points] Explain why the Mean Value Theorem applies to f on [0, 4].

(b) [2 points] Find the average rate of change of f on [0, 4].

(c) [3 points] Find all values c in (0, 4) guaranteed by the Mean Value Theorem.

(d) [2 points] Explain why the tangent line to the graph of f at t = c is parallel to the secant line through the endpoints.

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

Question 93: Absolute Extrema and the Candidates Test

★★★★★ VERY IMPORTANT 9 Points

Let $g(x) = x^3 - 6x^2 + 9x$ on the closed interval $[0, 5]$.

(a) [2 points] Find all interior critical numbers of g on [0, 5].

(b) [3 points] Evaluate g at the endpoints of the interval and at every interior critical number.

(c) [4 points] State the absolute maximum and absolute minimum values of g on [0, 5] and where they occur.

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

Question 94: Concavity and Points of Inflection

★★★★★ VERY IMPORTANT | 9 Points

Let f be the function given by $f(x) = x^4/4 - 2x^3/3$.

(a) [2 points] Find $f'(x)$ and $f''(x)$, and factor $f''(x)$ completely.

(b) [2 points] Determine the open intervals where the graph of f is concave up and concave down.

(c) [3 points] Find the coordinates of all points of inflection on the graph of f.

(d) [2 points] Explain why checking $f''(x) = 0$ alone is not sufficient to establish a point of inflection.

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

Question 95: Second Derivative Test

★★★★★ VERY IMPORTANT | 9 Points

Let $f(x) = x^4 - 4x^2$.

(a) [3 points] Find $f'(x)$, $f''(x)$, and find all critical numbers of f.

(b) [3 points] Use the Second Derivative Test to classify each critical number as a relative maximum, relative minimum, or inconclusive.

(c) [3 points] Find the coordinates of all relative extrema of f.

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

Question 96: Using a Sign Table for f′

★★★★★ VERY IMPORTANT 9 Points

The table describes a continuous, differentiable function f on the closed interval $[-2, 3]$.

x-interval	$(-2, -1)$	$(-1, 1)$	$(1, 3)$
sign of f′	+	−	+

x	−2	−1	1	3
f(x)	0	4	−2	1

(a) [3 points] State the open intervals where f is increasing and where f is decreasing.

(b) [2 points] Find and classify all relative extrema of f in the open interval $(-2, 3)$.

(c) [4 points] Determine the absolute maximum and absolute minimum values of f on $[-2, 3]$ and the x-values where they occur.

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

Question 97: Optimization: Fencing Beside a River

★★★★★ VERY IMPORTANT 9 Points

A farmer has 80 meters of fencing to enclose three sides of a rectangular corral adjacent to a straight river. The river bank forms the fourth side and requires no fencing. Let x be the length of each of the two sides perpendicular to the river, and let y be the side parallel to the river.

(a) [2 points] Write an equation for the fencing constraint and express the area A of the corral as a function of x alone, stating its physical domain.

(b) [4 points] Find the dimensions x and y that maximize the area of the corral.

(c) [3 points] Verify that the dimensions found in part (b) yield a maximum area, and state the maximum area.

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

Question 98: Optimization: Open-Top Storage Box

★★★★★ VERY IMPORTANT 9 Points

An open-top rectangular box is constructed by cutting square corners of side length x inches from a square sheet of cardboard measuring 12 inches by 12 inches and folding up the flaps.

(a) [2 points] Write an expression for the volume V(x) of the box in terms of x, and state the physical domain.

(b) [3 points] Find the critical point of V(x) within the physical domain.

(c) [4 points] Show that this critical point maximizes the volume, and state the dimensions of the box and the maximum volume.

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

Question 99: Behavior of an Implicit Curve

★★★★★ VERY IMPORTANT | 9 Points

Consider the upper semicircle defined by the equation $x^2 + y^2 = 25$ with $y \geq 0$ on the interval $-5 \leq x \leq 5$.

(a) [2 points] Use implicit differentiation to find dy/dx in terms of x and y.

(b) [3 points] Determine the intervals on which the upper semicircle is increasing and decreasing as a function of x.

(c) [2 points] Find the absolute maximum and absolute minimum values of y on [−5, 5] and where they occur.

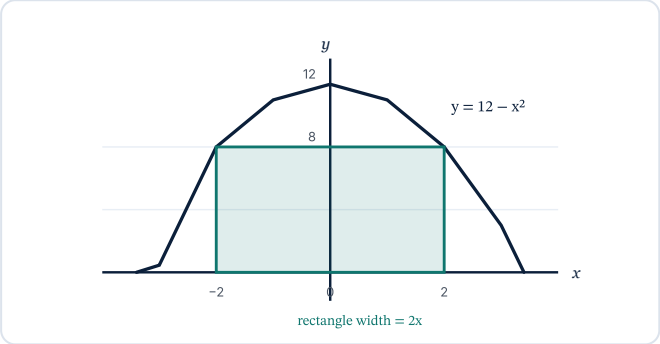
(d) [2 points] At which points does the curve have a vertical tangent? Justify your answer using dy/dx.

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

Question 100: Optimization: Inscribed Rectangle Under a Parabola

★★★★★ VERY IMPORTANT 9 Points

The diagram below shows a rectangle inscribed under the parabola $y = 12 - x^2$ and above the x-axis, with its base along the x-axis and upper vertices at $(-x, 12 - x^2)$ and $(x, 12 - x^2)$.



A symmetric rectangle is inscribed under $y = 12 - x^2$ and above the x-axis.

(a) [2 points] Express the area A of the rectangle as a function of x, and state the domain of x.

(b) [4 points] Find the critical value of x that maximizes the area of the rectangle.

(c) [3 points] Determine the dimensions (width and height) and the maximum area of the rectangle.

Reminder: Show all mathematical reasoning. Name any theorem (MVT, EVT) or test (1st/2nd Derivative Test) you use.

End of Question Paper

You have reached the end of Questions 1 through 100.

Before you continue: Make sure you have genuinely attempted every question. The pages that follow contain the complete answer key, fully worked multiple choice solutions with analysis of every incorrect option, and model free response solutions with AP-style scoring rubrics.

What Comes Next

- **Answer Key:** Letter answers for Questions 1–90 with priority ratings.
- **Multiple Choice Solutions:** Step-by-step reasoning for Questions 1–90 plus an explanation of why each distractor fails.
- **Free Response Solutions:** Complete model answers for Questions 91–100 with point-by-point rubrics.
- **Appendix:** Essential Unit 5 analytical theorems, derivative tests, and optimization strategies.

Multiple Choice Answer Key

Letter answers with priority star ratings. Questions 91–100 are free response and are scored using the rubrics that follow.

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
C ★★★	B ★★★	B ★★★	C ★★★	A ★★★	B ★★★	A ★★★	B ★★★	A ★★★★★	D ★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
B ★★★★★	D ★★★	B ★★★★★	B ★★★	A ★★★	D ★★★★★	D ★★★	B ★★★★★	B ★★★★★	B ★★★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
C ★★★★★	D ★★★★★	B ★★★★★	C ★★★	D ★★★	A ★★★	D ★★★	B ★★★★★	A ★★★	B ★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
A ★★★★★	A ★★★	D ★★★	C ★★★	D ★★★	B ★★	A ★★★★★	D ★★★	C ★★★	D ★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
A ★★★	B ★★★	B ★★★★★	A ★★★	C ★★★	B ★★★★★	B ★★★	C ★★★★★	D ★★★	D ★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
C ★★★★★	A ★★★★★	B ★★★★★	D ★★★	B ★★★★★	D ★★★	D ★★★	C ★★★★★	C ★★★★★	B ★★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
C ★★★	A ★★★	C ★★★★★	B ★★★★★	B ★★★★★	C ★★★★★	A ★★★★★	D ★★★	A ★★★	B ★★★★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
B ★★★★★	D ★★★★★	A ★★★★★	B ★★★	D ★★★★★	A ★★★★★	B ★★★★★	C ★★★★★	A ★★★	A ★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
C ★★★	B ★★★★★	B ★★★★★	B ★★★	C ★★★★★	C ★★★★★	B ★★★	D ★★★★★	D ★★★★★	C ★★★

Scoring Guide: Each multiple choice question is worth 1 point (90 total). Each free response question is worth 9 points (90 total). Maximum raw score: 180 points.

Question 01

★★★ MANDATORY

Correct Answer: (C) 2

STEP-BY-STEP SOLUTION

$f(0)=f(4)=0$, so the secant slope is 0. $f'(x)=2x-4$. Set $2c-4=0$ to get $c=2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The MVT point must be interior, not the endpoint 4.

✗ **(B)** At $c=1$, $f'(1)=-2$, not 0.

✗ **(D)** The MVT point must be in $(0,4)$, so 0 is not allowed.

Question 02

★★★ MANDATORY

Correct Answer: (B) Differentiability on (−1,1)

STEP-BY-STEP SOLUTION

The function $|x|$ is continuous on $[-1,1]$, but it has a corner at $x=0$ and is not differentiable there. The MVT therefore cannot be applied.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Continuity does hold on the closed interval.

✗ **(C)** Endpoint differentiability is not an MVT requirement.

✗ **(D)** The MVT does not require equal endpoint values.

Question 03

★★★ MANDATORY

Correct Answer: (B) 9/4

STEP-BY-STEP SOLUTION

The secant slope is $(2-1)/(4-1)=1/3$. Since $f'(x)=1/(2\sqrt{x})$, solve $1/(2\sqrt{c})=1/3$. Then $\sqrt{c}=3/2$ and $c=9/4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is the midpoint of the interval, but the MVT does not generally select the midpoint.

✗ **(C)** Substitution gives a value of c, not the derivative equation's solution.

✗ **(D)** This is an endpoint and is not in $(1,4)$.

Question 04

★★★ MANDATORY

Correct Answer: (C) $2/\sqrt{3}$

STEP-BY-STEP SOLUTION

The secant slope is $(8-0)/(2-0)=4$. Since $f'(x)=3x^2$, set $3c^2=4$. The interior solution is $c=2/\sqrt{3}$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Substitution gives $3c^2=1$, not 4.

✗ **(B)** This is not the solution of $3c^2=4$.

✗ **(D)** This does not solve $3c^2=4$.

Question 05

★★★ MANDATORY

Correct Answer: (A) $\sqrt{2}$

STEP-BY-STEP SOLUTION

The secant slope is $(1/2-1)/(2-1)=-1/2$. Since $f'(x)=-1/x^2$, solve $-1/c^2=-1/2$, giving $c=\sqrt{2}$, which lies in $(1,2)$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** This is outside $(1,2)$.

✗ **(C)** The midpoint does not satisfy $f'(c)=-1/2$.

✗ **(D)** An endpoint cannot be the MVT value.

Question 06

★★★ MANDATORY

Correct Answer: (B) $c=3$

STEP-BY-STEP SOLUTION

$f(2)=0$ and $f(4)=0$; the polynomial is continuous and differentiable. $f'(x)=2x-6=0$ gives $c=3$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** An endpoint is not the required interior value.

✗ **(C)** An endpoint is not the required interior value.

✗ **(D)** All Rolle's Theorem hypotheses hold, and $c=3$ works.

Question 07

★★★ MANDATORY

Correct Answer: (A) e−1

STEP-BY-STEP SOLUTION

The secant slope is (ln e−ln 1)/(e−1)=1/(e−1). Since f′(x)=1/x, set 1/c=1/(e−1), giving c=e−1. This is in (1,e).

WHY THE OTHER OPTIONS FAIL

✗ **(B)** e is an endpoint, not an interior point.

✗ **(C)** This is the secant slope, not the input c.

✗ **(D)** There is no reason the MVT point must be the geometric mean.

Question 08

★★★ MANDATORY

Correct Answer: (B) π/2

STEP-BY-STEP SOLUTION

The endpoint values are both 0, so the secant slope is 0. f′(x)=cos x; cos c=0 gives c=π/2 in (0,π).

WHY THE OTHER OPTIONS FAIL

✗ **(A)** π is an endpoint, not an interior point.

✗ **(C)** cos(π/4) is not 0.

✗ **(D)** cos(3π/4) is not 0.

Question 09

★★★★★ VERY IMPORTANT

Correct Answer: (A) f has an absolute maximum and an absolute minimum on [a,b]

STEP-BY-STEP SOLUTION

The Extreme Value Theorem states that a function continuous on a closed, bounded interval attains both an absolute maximum and an absolute minimum there.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The theorem guarantees extrema, not a particular number of critical numbers.

✗ **(C)** Continuity gives no monotonicity guarantee.

✗ **(D)** Continuity does not imply differentiability.

Question 10

★★★ MANDATORY

Correct Answer: (D) neither an absolute maximum nor an absolute minimum

STEP-BY-STEP SOLUTION

The values approach 0 and 1, but neither endpoint is included. Thus the infimum and supremum are not attained; there is no absolute minimum or maximum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The infimum 0 is not attained.
✗ **(B)** Neither endpoint value is attained.
✗ **(C)** The supremum 1 is not attained.

Question 11

★★★★★ VERY IMPORTANT

Correct Answer: (B) f is continuous on the closed interval [a,b]

STEP-BY-STEP SOLUTION

The EVT applies to every function continuous on a closed, bounded interval. The other statements are not required.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Equal endpoint values are not required.
✗ **(C)** Monotonicity is not required.
✗ **(D)** Many continuous nonpolynomial functions satisfy EVT.

Question 12

★★★ MANDATORY

Correct Answer: (D) two solutions $c=\pm 1/\sqrt{3}$

STEP-BY-STEP SOLUTION

The secant slope is $(1-(-1))/2=1$. Since $f'(x)=3x^2$, solve $3c^2=1$, giving $c=\pm 1/\sqrt{3}$; both lie in $(-1,1)$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The negative solution also satisfies the equation.
✗ **(B)** The derivative at 0 is 0, not 1.
✗ **(C)** Both values satisfy the MVT equation.

Question 13

★★★★★ VERY IMPORTANT

Correct Answer: (B) (0,2)

STEP-BY-STEP SOLUTION

Check the sign of x(x−2): it is positive for x<0, negative for 02. Therefore f decreases on (0,2).

WHY THE OTHER OPTIONS FAIL

✗ **(A)** f′ is positive there, so f increases.

✗ **(C)** f′ is positive there, so f increases.

✗ **(D)** These are intervals of increase.

Question 14

★★★ MANDATORY

Correct Answer: (B) f increases on all real numbers

STEP-BY-STEP SOLUTION

$f'(x) \geq 0$ and is positive except at $x=1$. For example, an antiderivative is $(x-1)^3$, which is strictly increasing; the isolated zero does not create an extremum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** f′ does not change from negative to positive at 1.

✗ **(C)** The derivative is not negative on intervals.

✗ **(D)** f′ does not change from positive to negative at 1.

Question 15

★★★ MANDATORY

Correct Answer: (A) f is strictly decreasing on the real line

STEP-BY-STEP SOLUTION

$f' \leq 0$ and is negative except at $x=-2$. The derivative does not change sign there, so f remains strictly decreasing and has no local extremum at −2.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The derivative is not positive on any interval.

✗ **(C)** There is no positive-to-negative sign change.

✗ **(D)** There is no negative-to-positive sign change.

Question 16

★★★★★ VERY IMPORTANT

Correct Answer: (D) a local maximum at −1 and minimum at 3

STEP-BY-STEP SOLUTION

At −1, f′ changes positive to negative, so f has a local maximum. At 3, f′ changes negative to positive, so f has a local minimum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Reverses both sign-change conclusions.
✗ **(B)** The sign changes are different at the two points.
✗ **(C)** The sign changes are different at the two points.

Question 17

★★★ MANDATORY

Correct Answer: (D) f′(0)=0, but there is no local extremum

STEP-BY-STEP SOLUTION

$f'(x)=3x^2$, so $f'(0)=0$. But f′ is positive on both sides of 0, so f increases through 0 and has no local extremum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** f′ is positive on both sides, so f is increasing.
✗ **(B)** f increases on both sides, so no minimum.
✗ **(C)** f increases on both sides, so no maximum.

Question 18

★★★★★ VERY IMPORTANT

Correct Answer: (B) a local minimum at 2

STEP-BY-STEP SOLUTION

The First Derivative Test says a change from negative to positive means f changes from decreasing to increasing, giving a local minimum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The sign of f′ alone does not determine concavity.
✗ **(C)** The sign change identifies a critical point.
✗ **(D)** A maximum requires positive-to-negative change.

Question 19

★★★★★ VERY IMPORTANT

Correct Answer: (B) f has a local minimum at x=−1 and no extremum at x=2

STEP-BY-STEP SOLUTION

The squared factor is nonnegative, so the sign of f′ is determined by x+1. It changes negative to positive at −1, giving a local minimum. It does not change sign at 2, so there is no extremum there.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** There is no sign change at x=2.

✗ **(C)** The sign changes − to + at −1, not + to −; x=2 has no sign change.

✗ **(D)** There is no sign change at x=2.

Question 20

★★★★★ VERY IMPORTANT

Correct Answer: (B) c is in the domain of f and f′(c)=0 or f′(c) is undefined

STEP-BY-STEP SOLUTION

A critical number is an input in the domain of f where f′ is zero or does not exist.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** A zero of f is an x-intercept, not necessarily a critical number.

✗ **(C)** This omits points where the derivative equals zero.

✗ **(D)** This omits points where the derivative is undefined.

Question 21

★★★★★ VERY IMPORTANT

Correct Answer: (C) f′(c)=0

STEP-BY-STEP SOLUTION

Fermat's Theorem states that if a differentiable function has a local extremum at an interior point, then its derivative there is zero.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The second derivative need not be zero.

✗ **(B)** Differentiability at c is given.

✗ **(D)** An extremum need not have value zero.

Question 22

★★★★★ VERY IMPORTANT

Correct Answer: (D) f has a local maximum at c

STEP-BY-STEP SOLUTION

Positive f′ means increasing; negative f′ means decreasing. The transition is increasing to decreasing, so c is a local maximum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Negative-to-positive gives a local minimum.
✗ **(B)** The function changes direction, so it is not constant.
✗ **(C)** An inflection point requires a concavity change.

Question 23

★★★★★ VERY IMPORTANT

Correct Answer: (B) f has a local minimum at c

STEP-BY-STEP SOLUTION

Negative f′ means decreasing; positive f′ means increasing. The transition gives a local minimum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Positive-to-negative gives a maximum.
✗ **(C)** The derivative signs are different on the two sides.
✗ **(D)** A sign change through zero implies a critical number.

Question 24

★★★ MANDATORY

Correct Answer: (C) $f(x)=x^3$ at $c=0$

STEP-BY-STEP SOLUTION

For $f(x)=x^3$, $f'(0)=0$, but the function is increasing on both sides of 0 and has no local extremum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** x^2 has a local minimum at 0.
✗ **(B)** $-x^2$ has a local maximum at 0.
✗ **(D)** The derivative is undefined at 0, not zero.

Question 25

★★★ MANDATORY

Correct Answer: (D) f is defined at 1 but f′(1) does not exist

STEP-BY-STEP SOLUTION

The absolute-value graph is continuous and defined at 1, but has a corner there, so its derivative does not exist. That makes 1 a critical number.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The absolute-value function is continuous at 1.
✗ **(B)** The derivative at the corner does not exist.
✗ **(C)** The function value alone does not define a critical number.

Question 26

★★★ MANDATORY

Correct Answer: (A) increasing on that interval

STEP-BY-STEP SOLUTION

A positive first derivative means the function is increasing.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** A constant function has derivative zero.
✗ **(C)** Concavity is determined by f″ or the behavior of f′.
✗ **(D)** A negative derivative means decreasing.

Question 27

★★★ MANDATORY

Correct Answer: (D) decreasing on (a,b)

STEP-BY-STEP SOLUTION

A negative first derivative means the function is decreasing.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** A positive derivative means increasing.
✗ **(B)** Concavity is not determined by the sign of f′ alone.
✗ **(C)** A constant function has derivative zero.

Question 28

★★★★★ VERY IMPORTANT

Correct Answer: (B) minimum at 2 and maximum at 5

STEP-BY-STEP SOLUTION

At 2 the sign changes − to +, giving a local minimum. At 5 it changes + to −, giving a local maximum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The sign changes at the two points are opposite.

✗ **(C)** Reverses both sign-change conclusions.

✗ **(D)** The sign changes at the two points are opposite.

Question 29

★★★ MANDATORY

Correct Answer: (A) continues decreasing through a stationary point

STEP-BY-STEP SOLUTION

f′ is nonpositive and negative on both sides of 3. The derivative is zero at 3 but does not change sign, so the function continues decreasing.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** There is no + to − sign change.

✗ **(C)** There is no − to + sign change.

✗ **(D)** The derivative is negative on both sides.

Question 30

★★★ MANDATORY

Correct Answer: (B) the function is defined there and its nearby values support an extremum

STEP-BY-STEP SOLUTION

A cusp such as $f(x)=|x|$ has a local minimum at 0 even though $f'(0)$ does not exist; the function itself is defined there.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Positive derivative on both sides indicates increase, not an extremum.

✗ **(C)** This is not the criterion for an extremum.

✗ **(D)** A point outside the domain cannot be a local extremum of the function.

Question 31

★★★★★ VERY IMPORTANT

Correct Answer: (A) endpoints and interior critical numbers

STEP-BY-STEP SOLUTION

Evaluate f at both endpoints and at every interior critical number (where $f'=0$ or is undefined) in the domain, then compare the values.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Interior critical numbers and endpoints must both be checked.

✗ **(C)** Inflection points are not the only candidates and need not be extrema.

✗ **(D)** Interior critical numbers may contain extrema.

Question 32

★★★ MANDATORY

Correct Answer: (A) 2

STEP-BY-STEP SOLUTION

$f'=3x^2-3=0$ at $x=\pm1$. Values: $f(-2)=-2$, $f(-1)=2$, $f(1)=-2$, $f(2)=2$. The maximum is 2.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** This is a minimum value.

✗ **(C)** No candidate point has value 4.

✗ **(D)** This is not attained as the maximum.

Question 33

★★★ MANDATORY

Correct Answer: (D) −1

STEP-BY-STEP SOLUTION

$f'=2x-2=0$ at $x=1$. Candidate values: $f(0)=0$, $f(1)=-1$, $f(3)=3$. The minimum is -1.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is not the minimum candidate value.

✗ **(B)** This is the value at $x=0$, not the minimum.

✗ **(C)** This is the value at the right endpoint.

Question 34

★★★ MANDATORY

Correct Answer: (C) −4

STEP-BY-STEP SOLUTION

$f'=3x(x-2)$, so the interior critical number is $x=2$. Values at 0, 2, 3 are 0, −4, 0. The absolute minimum is −4.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is the endpoint value, not the minimum.

✗ **(B)** This is not attained as the minimum.

✗ **(D)** No candidate gives −3.

Question 35

★★★ MANDATORY

Correct Answer: (D) minimum 0, maximum 2

STEP-BY-STEP SOLUTION

The smallest value is $f(0)=0$. Endpoint values are $f(-1)=1$ and $f(2)=2$, so the greatest value is 2.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Absolute value is never negative.

✗ **(B)** Both proposed values are incorrect.

✗ **(C)** Misses the endpoint value 2.

Question 36

★★ IMPORTANT

Correct Answer: (B) x=−2

STEP-BY-STEP SOLUTION

Candidate values: $f(-2)=4$, $f(0)=0$, $f(1)=1$. The maximum is 4 at $x=-2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Neither point gives the maximum 4.

✗ **(C)** This gives the minimum.

✗ **(D)** This value is 1.

Question 37

★★★★★ VERY IMPORTANT

Correct Answer: (A) attain both an absolute maximum and an absolute minimum

STEP-BY-STEP SOLUTION

The EVT guarantees an attained absolute maximum and minimum on a closed, bounded interval.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Continuity does not imply differentiability.

✗ **(C)** There may be none or several local maxima.

✗ **(D)** The extrema could occur only at the endpoints.

Question 38

★★★ MANDATORY

Correct Answer: (D) a and b

STEP-BY-STEP SOLUTION

Strict increase means f(a) is the smallest value and f(b) is the largest value.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Reverses the order.

✗ **(B)** f(b)>f(a), so both cannot occur at a.

✗ **(C)** No interior critical point is required.

Question 39

★★★ MANDATORY

Correct Answer: (C) 1

STEP-BY-STEP SOLUTION

$f'=-2x+2=0$ at $x=1$. Values: $f(0)=0$, $f(1)=1$, $f(3)=-3$. The maximum is 1.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The right endpoint value is −3, not 3.

✗ **(B)** This is only the left endpoint value.

✗ **(D)** This is the minimum.

Question 40

★★★ MANDATORY

Correct Answer: (D) 5

STEP-BY-STEP SOLUTION

$f'=2x-4=0$ at $x=2$. Candidate values: $f(0)=0$, $f(2)=-4$, $f(5)=5$. The maximum is 5.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is the left endpoint value.

✗ **(B)** Not a candidate value.

✗ **(C)** This is the minimum.

Question 41

★★★ MANDATORY

Correct Answer: (A) −5

STEP-BY-STEP SOLUTION

$f'=-2x+4=0$ at $x=2$. Values: $f(0)=0$, $f(2)=4$, $f(5)=-5$. The minimum is -5.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** This is the left endpoint value.

✗ **(C)** This is the interior maximum.

✗ **(D)** This is not the minimum.

Question 42

★★★ MANDATORY

Correct Answer: (B) neither an absolute maximum nor an absolute minimum

STEP-BY-STEP SOLUTION

The values approach 0 and 1 but never equal either value because the endpoints are excluded.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** 1 is a supremum but is not attained.

✗ **(C)** 0 is an infimum but is not attained.

✗ **(D)** Neither endpoint value is attained.

Question 43

★★★★★ VERY IMPORTANT

Correct Answer: (B) an absolute maximum and an absolute minimum, not necessarily unique

STEP-BY-STEP SOLUTION

The EVT guarantees existence of attained absolute extrema, but they need not be unique and may occur at interior points.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The maximum can occur in the interior.
✗ **(C)** Differentiability is not guaranteed.
✗ **(D)** The maximum can occur at multiple inputs.

Question 44

★★★ MANDATORY

Correct Answer: (A) 0 at x=0

STEP-BY-STEP SOLUTION

$x^{(2/3)}=|x|^{(2/3)}\geq 0$, equals 0 at $x=0$, and equals 1 at both endpoints.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The function is nonnegative.
✗ **(C)** These are endpoint values and are not the minimum.
✗ **(D)** The continuous function attains a minimum on the closed interval.

Question 45

★★★ MANDATORY

Correct Answer: (C) 1/5 at x=−2

STEP-BY-STEP SOLUTION

$f'=-2x/(1+x^2)^2$, so the critical number is $x=0$. Candidate values: $f(-2)=1/5$, $f(0)=1$, $f(1)=1/2$. The minimum is $1/5$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is the maximum.
✗ **(B)** This is an endpoint value, but it is larger than 1/5.
✗ **(D)** The function is positive on the interval.

Question 46

★★★★★ VERY IMPORTANT

Correct Answer: (B) must be checked if it belongs to the domain of f

STEP-BY-STEP SOLUTION

Interior domain points where f′ does not exist are critical numbers and must be included among the candidates, along with endpoints.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Such a point can be a cusp extremum.

✗ **(C)** For example, $|x|$ has a minimum at a nondifferentiable point.

✗ **(D)** Undefined derivative does not automatically imply a maximum.

Question 47

★★★ MANDATORY

Correct Answer: (B) $(-\infty,0)$

STEP-BY-STEP SOLUTION

Concave down where $f''<0$. Since $6x<0$ when $x<0$, f is concave down on $(-\infty,0)$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The sign changes at 0.

✗ **(C)** There is an interval where $f''<0$.

✗ **(D)** $f''>0$ for $x>0$, so the graph is concave up there.

Question 48

★★★★★ VERY IMPORTANT

Correct Answer: (C) There is no change in concavity at 1

STEP-BY-STEP SOLUTION

f″ is positive on both sides of 1 and only equals zero at 1. Since concavity does not change, x=1 is not an inflection point based on this information.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** An inflection point requires a change in concavity.

✗ **(B)** f″ alone does not show a local maximum at 1.

✗ **(D)** f″ alone does not show a local minimum at 1.

Question 49

★★★ MANDATORY

Correct Answer: (D) $(-\infty,-2)\cup(2,\infty)$

STEP-BY-STEP SOLUTION

$f''=x^2-4$ is positive when $|x|>2$ and negative when $|x|<2$. Thus the graph is concave up outside $[-2,2]$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** It includes values where $f''<0$.

✗ **(B)** It includes values where $f''<0$.

✗ **(C)** This is the interval of concave down.

Question 50

★★★ MANDATORY

Correct Answer: (D) $x=-2$ and $x=1$

STEP-BY-STEP SOLUTION

The possible changes occur at zeros -2 and 1. The product is positive on $(-\infty,-2)$, negative on $(-2,1)$, and positive on $(1,\infty)$, so concavity changes at both.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** These are not the zeros of f''.

✗ **(B)** Concavity also changes at $x=1$.

✗ **(C)** Concavity also changes at $x=-2$.

Question 51

★★★★★ VERY IMPORTANT

Correct Answer: (C) a local minimum

STEP-BY-STEP SOLUTION

A positive second derivative means the graph is concave up at c. With $f'(c)=0$, the Second Derivative Test identifies a local minimum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** An inflection point requires a concavity change.

✗ **(B)** $f''<0$ would indicate a local maximum.

✗ **(D)** The test is conclusive when $f''(c)>0$.

Question 52

★★★★★ VERY IMPORTANT

Correct Answer: (A) a local maximum

STEP-BY-STEP SOLUTION

With $f'(c)=0$ and $f''(c)<0$, the graph is concave down at the critical point, so c is a local maximum.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The test is conclusive when $f''(c)<0$.

✗ **(C)** A concavity change is not implied.

✗ **(D)** That corresponds to $f''>0$.

Question 53

★★★★★ VERY IMPORTANT

Correct Answer: (B) The test is inconclusive; further analysis shows a local minimum

STEP-BY-STEP SOLUTION

Since $f''(0)=0$, the Second Derivative Test is inconclusive. But $x^4 \geq 0$ with equality at 0, so direct analysis shows a local minimum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The test itself does not decide when $f''=0$.

✗ **(C)** Concavity is up on both sides of 0.

✗ **(D)** x^4 is not less than nearby values.

Question 54

★★★ MANDATORY

Correct Answer: (D) neither a local maximum nor a local minimum

STEP-BY-STEP SOLUTION

$f(x)=x^3$ is increasing through 0. Values to the left are smaller and values to the right are larger, so there is no local extremum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The function is not greater than nearby values on both sides.

✗ **(B)** The function is not less than nearby values on both sides.

✗ **(C)** The function is unbounded on the real line.

Question 55

★★★★★ VERY IMPORTANT

Correct Answer: (B) the concavity of f changes at c and the point lies on the graph

STEP-BY-STEP SOLUTION

An inflection point is a point on the graph where concavity changes. $f''(c)=0$ is a possible clue, not a sufficient definition.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** f'' can be zero without a concavity change.

✗ **(C)** A local extremum need not be an inflection point.

✗ **(D)** A horizontal tangent is not required.

Question 56

★★★ MANDATORY

Correct Answer: (D) increasing

STEP-BY-STEP SOLUTION

A positive second derivative means the first derivative is increasing, and f is concave up.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** A positive f'' is incompatible with f' identically zero.

✗ **(B)** A constant f' would give $f''=0$.

✗ **(C)** That corresponds to $f''<0$.

Question 57

★★★ MANDATORY

Correct Answer: (D) concave down and f′ is decreasing

STEP-BY-STEP SOLUTION

Negative second derivative means the graph is concave down and its first derivative is decreasing.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Both descriptions reverse the sign.

✗ **(B)** A constant function has $f''=0$.

✗ **(C)** The sign of f' is not given.

Question 58

★★★★★ VERY IMPORTANT

Correct Answer: (C) f has an inflection point but no local extremum

STEP-BY-STEP SOLUTION

$f'(0)=0$, but $f'=3x^2$ is positive on both sides, so no local extremum occurs. $f''=6x$ changes from negative to positive, so concavity changes and $x=0$ is an inflection point.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The function increases through 0.

✗ **(B)** Concavity changes at 0.

✗ **(D)** The function increases through 0.

Question 59

★★★★★ VERY IMPORTANT

Correct Answer: (C) x=0 is not an inflection point because concavity is upward on both sides

STEP-BY-STEP SOLUTION

$f''(x)=12x^2$ is positive for $x\neq 0$ on both sides of 0. Concavity does not change, so 0 is not an inflection point; it is a local minimum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The function decreases to 0 and increases after 0.

✗ **(B)** $f''=0$ alone is not sufficient.

✗ **(D)** $x^4\geq 0$, so 0 is a minimum, not a maximum.

Question 60

★★★ MANDATORY

Correct Answer: (B) −1 and 2

STEP-BY-STEP SOLUTION

Concavity changes at −1 from down to up and at 2 from up to down.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Concavity also changes at −1.

✗ **(C)** Zero is not a stated sign-change point.

✗ **(D)** Concavity also changes at 2.

Question 61

★★★ MANDATORY

Correct Answer: (C) concave up there

STEP-BY-STEP SOLUTION

An increasing first derivative corresponds to positive second derivative and concave-up behavior.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** A decreasing f′ corresponds to concave down.

✗ **(B)** f′ can be increasing while positive or negative; f's monotonicity is not fixed.

✗ **(D)** A constant function has constant f′.

Question 62

★★★ MANDATORY

Correct Answer: (A) concave down there

STEP-BY-STEP SOLUTION

A decreasing first derivative corresponds to negative second derivative and concave-down behavior.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The sign of f′ is not specified.

✗ **(C)** That corresponds to increasing f′.

✗ **(D)** The sign of f′ is not specified.

Question 63

★★★★★ VERY IMPORTANT

Correct Answer: (C) f is continuous and its concavity changes across the point

STEP-BY-STEP SOLUTION

For example, f(x)=x^(1/3) is continuous at 0, f″ is undefined there, and concavity changes across 0. Thus f″ need not exist at an inflection point.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** A discontinuity is not an inflection point on the graph.

✗ **(B)** The sign of f′ does not determine concavity.

✗ **(D)** The question specifically allows f″ to be undefined.

Question 64

★★★★★ VERY IMPORTANT

Correct Answer: (B) The Second Derivative Test is inconclusive

STEP-BY-STEP SOLUTION

When $f''(c)=0$, the Second Derivative Test gives no conclusion. A separate sign analysis or other argument is needed.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** A concavity change is not guaranteed.
✗ **(C)** The test cannot guarantee a minimum.
✗ **(D)** The test cannot guarantee a maximum.

Question 65

★★★★★ VERY IMPORTANT

Correct Answer: (B) increasing and concave down

STEP-BY-STEP SOLUTION

$f'>0$ means f is increasing. $f''<0$ means f is concave down.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** f' is positive, so f is not decreasing.
✗ **(C)** Both signs are inconsistent with this choice.
✗ **(D)** f'' is negative, so it is not concave up.

Question 66

★★★★★ VERY IMPORTANT

Correct Answer: (C) a local maximum at c

STEP-BY-STEP SOLUTION

The function changes from increasing to decreasing, so c is a local maximum by the First Derivative Test.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** No tangent behavior is specified.
✗ **(B)** A sign change of f' alone does not establish an inflection point.
✗ **(D)** Negative-to-positive gives a local minimum.

Question 67

★★★★★ VERY IMPORTANT

Correct Answer: (A) a local minimum at c

STEP-BY-STEP SOLUTION

The function changes from decreasing to increasing, so c is a local minimum.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The sign of f″ is not determined by this change alone.

✗ **(C)** Positive-to-negative gives a local maximum.

✗ **(D)** An extremum of a differentiable function is not a vertical asymptote.

Question 68

★★★ MANDATORY

Correct Answer: (D) does not have a local extremum at c by the First Derivative Test

STEP-BY-STEP SOLUTION

No sign change in f′ means f keeps the same increasing/decreasing behavior through c, so the First Derivative Test gives no local extremum there.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** A concavity change is not determined.

✗ **(B)** A maximum needs a + to − change.

✗ **(C)** A minimum needs a − to + change.

Question 69

★★★ MANDATORY

Correct Answer: (A) decreasing and concave up

STEP-BY-STEP SOLUTION

The negative first derivative means f decreases; the positive second derivative means f is concave up.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** f′ is negative, not positive.

✗ **(C)** Both monotonicity and concavity are misread.

✗ **(D)** f″ is positive, not negative.

Question 70

★★★★★ VERY IMPORTANT

Correct Answer: (B) a local maximum at 4

STEP-BY-STEP SOLUTION

Above the x-axis f′>0 and below it f′<0. Thus f changes from increasing to decreasing and has a local maximum at 4.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The sign of f′ alone does not establish concavity change.
✗ **(C)** The zero of f′ is a critical number.
✗ **(D)** A minimum requires below-to-above crossing.

Question 71

★★★★★ VERY IMPORTANT

Correct Answer: (B) increasing and concave up

STEP-BY-STEP SOLUTION

Positive f′ means f increases. Increasing f′ means f is concave up.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** An increasing f′ gives concave up, not down.
✗ **(C)** Positive f′ means increasing.
✗ **(D)** Both conclusions are reversed.

Question 72

★★★★★ VERY IMPORTANT

Correct Answer: (D) decreasing and concave down

STEP-BY-STEP SOLUTION

Negative f′ means f decreases. Decreasing f′ means f is concave down.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Both signs conflict with the information.
✗ **(B)** A decreasing f′ means concave down.
✗ **(C)** Negative f′ means decreasing.

Question 73

★★★★★ VERY IMPORTANT

Correct Answer: (A) 100

STEP-BY-STEP SOLUTION

Let sides be x and y. $x+y=20$, so $A=x(20-x)=-(x-10)^2+100$. The maximum area is 100 at $x=y=10$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The maximum is larger than 80.

✗ **(C)** 400 is not compatible with perimeter 40.

✗ **(D)** The vertex value is 100.

Question 74

★★★ MANDATORY

Correct Answer: (B) 225

STEP-BY-STEP SOLUTION

Let the numbers be x and $30-x$. $P=x(30-x)$, a downward parabola with vertex $x=15$. The maximum product is $15 \cdot 15=225$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The maximum occurs at equality, giving 225.

✗ **(C)** The product can be increased by moving closer to equal numbers.

✗ **(D)** No pair with sum 30 has product 240.

Question 75

★★★★★ VERY IMPORTANT

Correct Answer: (D) 800 m^2

STEP-BY-STEP SOLUTION

Let x be each perpendicular side and y the fenced side parallel to the river. $2x+y=80$, so $A=x(80-2x)=80x-2x^2$. The vertex is $x=20$, $y=40$, giving $A=800$ m^2.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The vertex calculation gives a larger area.

✗ **(B)** This exceeds the maximum of the area quadratic.

✗ **(C)** At $x=20$ the area is 800, and the quadratic has no larger value.

Question 76

★★★★★ VERY IMPORTANT

Correct Answer: (A) 2

STEP-BY-STEP SOLUTION

$V=x(12-2x)^2$ for 0

WHY THE OTHER OPTIONS FAIL

✗ **(B)** V' is positive at $x=1$.

✗ **(C)** V' is negative at $x=3$, after the maximum.

✗ **(D)** At $x=6$ the box has zero base dimensions.

Question 77

★★★★★ VERY IMPORTANT

Correct Answer: (B) $12\sqrt{3}$

STEP-BY-STEP SOLUTION

The height is $9-x^2$ and width is $2x$, so $A=2x(9-x^2)$, $0\le x\le 3$. $A'=18-6x^2=0$ gives $x=\sqrt{3}$; height$=6$. Thus $A=2\sqrt{3}\cdot 6=12\sqrt{3}$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is not the product of the maximizing width and height.

✗ **(C)** This is not the area at the critical point.

✗ **(D)** The height at $x=\sqrt{3}$ is 6, not 9.

Question 78

★★★★★ VERY IMPORTANT

Correct Answer: (C) height equal to twice its radius

STEP-BY-STEP SOLUTION

For fixed $V=\pi r^2 h$, $h=V/(\pi r^2)$. Surface area $S=2\pi r^2+2\pi rh=2\pi r^2+2V/r$. Setting $S'=4\pi r-2V/r^2=0$ gives $V=2\pi r^3$. Substituting into $h=V/(\pi r^2)$ gives $h=2r$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is the reciprocal of the required ratio.

✗ **(B)** The derivative condition gives $h=2r$.

✗ **(D)** This does not satisfy the minimizing condition.

Question 79

★★★ MANDATORY

Correct Answer: (A) 12

STEP-BY-STEP SOLUTION

By AM–GM, $x+y \geq 2\sqrt{(xy)}=2\sqrt{36}=12$, with equality at $x=y=6$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** A larger value is possible but not minimal.

✗ **(C)** This is the product, not the minimum sum.

✗ **(D)** The sum cannot be below 12.

Question 80

★★★ MANDATORY

Correct Answer: (A) 32

STEP-BY-STEP SOLUTION

For fixed area, the rectangle with minimum perimeter is a square. Side length $\sqrt{64}=8$, so perimeter $=4 \cdot 8=32$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** This is less than the perimeter of any 64-area rectangle.

✗ **(C)** The square's perimeter is 32.

✗ **(D)** This is the perimeter of a 6-by-6 square, whose area is 36.

Question 81

★★★ MANDATORY

Correct Answer: (C) x=15, y=7.5

STEP-BY-STEP SOLUTION

$y=(30-x)/2$, so $P=xy=(30x-x^2)/2$. $P'=15-x=0$ gives $x=15$ and $y=7.5$, an interior maximum since $P''=-1$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The problem requires $y>0$, and this gives zero product.

✗ **(B)** These values do not satisfy the constraint.

✗ **(D)** These values do not satisfy $x+2y=30$.

Question 82

★★★★★ VERY IMPORTANT

Correct Answer: (B) 50

STEP-BY-STEP SOLUTION

Define half-width u and half-height v: $u^2+v^2=25$ and area=4uv. Since $2uv \le u^2+v^2=25$, $4uv \le 50$. Maximum occurs when $u=v=5/\sqrt{2}$, giving maximum area 50.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is not the maximum value for the full rectangle.

✗ **(C)** This is below the attainable maximum.

✗ **(D)** This omits the factor relating half-dimensions to full area.

Question 83

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\pm\sqrt{(5/2)}$

STEP-BY-STEP SOLUTION

Minimize squared distance $D^2=x^2+(x^2-3)^2$. Its derivative is $2x+4x(x^2-3)=2x(2x^2-5)$. Critical points are 0 and $\pm\sqrt{(5/2)}$. At 0, $D^2=9$; at $\pm\sqrt{(5/2)}$, $D^2=11/4$, so the two latter points are closest.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** At x=0 the squared distance is 9, not the minimum.

✗ **(C)** These values do not solve $2x^2-5=0$.

✗ **(D)** This omits the symmetric point on the other side.

Question 84

★★★ MANDATORY

Correct Answer: (B) 20

STEP-BY-STEP SOLUTION

$P'(x)=-2x+40=0$ gives x=20. Since $P''=-2<0$, this is a maximum; checking the endpoints confirms it is absolute.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The right endpoint is not the maximizing critical point.

✗ **(C)** $P'(10)=20$, not zero.

✗ **(D)** $P'(40)=-40$, not zero.

Question 85

★★★★★ VERY IMPORTANT

Correct Answer: (C) 64

STEP-BY-STEP SOLUTION

$y=12-2x$, so $F=x^2(12-2x)$, 0

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The constraint gives $y=4$ when $x=4$, not 6.

✗ **(B)** Substitution at the critical point gives 64.

✗ **(D)** The value at $x=2$ is 32, not the maximum.

Question 86

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2,500/3 m²

STEP-BY-STEP SOLUTION

Let x be the perpendicular side length. The two outer perpendicular sides and divider use 3x; the far side has length y. Thus $3x+y=100$ and $A=x(100-3x)$. $A'=100-6x=0$ gives $x=50/3$ and $y=50$. Area $=(50/3)(50)=2500/3$ m².

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is not the maximum of the area quadratic.

✗ **(B)** The maximizing area is 2500/3, approximately 833.3.

✗ **(D)** This is not the product at the maximizing dimensions.

Question 87

★★★ MANDATORY

Correct Answer: (B) x=30

STEP-BY-STEP SOLUTION

Revenue $R=xp=120x-2x^2$. $R'=120-4x=0$ gives $x=30$, and $R''=-4<0$, so revenue is maximized there.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** $R'(40)<0$, after the maximum.

✗ **(C)** At $x=60$ the price is zero.

✗ **(D)** $R'(20)>0$, so revenue is still increasing.

Question 88

★★★★★ VERY IMPORTANT

Correct Answer: (D) x=0 or an endpoint, so values must be compared

STEP-BY-STEP SOLUTION

The sign chart gives a local maximum at x=0 and a local minimum at x=3. For an absolute maximum on a closed interval, compare f(−2), f(0), f(3), and f(4); the sign chart alone does not select the largest value.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The endpoint may be smaller than f(0) or f(4).

✗ **(B)** The endpoint may be smaller than f(0).

✗ **(C)** x=3 is a local minimum by the sign chart.

Question 89

★★★★★ VERY IMPORTANT

Correct Answer: (D) $10\sqrt{2}$ m

STEP-BY-STEP SOLUTION

Let x be each perpendicular side and y the side parallel to the wall. xy=400, so fencing $L=2x+y=2x+400/x$. $L'=2-400/x^2=0$ gives $x^2=200$, hence $x=10\sqrt{2}$ m; $L''=800/x^3>0$ confirms a minimum.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is the parallel side length at the minimum, not a perpendicular side.

✗ **(B)** At x=10, L is not minimized.

✗ **(C)** This is not the solution of $x^2=200$.

Question 90

★★★ MANDATORY

Correct Answer: (C) 36

STEP-BY-STEP SOLUTION

$A=xy=x(12-x)=12x-x^2$ on [0,12]. $A'=12-2x=0$ gives x=6 and y=6; A=36.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The vertex value is larger than 30.

✗ **(B)** No feasible x gives area 48.

✗ **(D)** The vertex value is larger than 24.

Question 91: Analyzing f′ from a Graph

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) The graph of f′ is above the axis for 0<t<3 and 7<t<8, so f increases on **(0, 3) and (7, 8)**. It is below the axis for 3<t<7, so f decreases on **(3, 7)**.

(b) f′ changes from + to − at t=3, so f has a **relative maximum at t = 3**. It changes from − to + at t=7, so f has a **relative minimum at t = 7**.

(c) Signed areas: [0, 2] = +4; [2, 3] = +1; [3, 4] = −1; [4, 6] = −4; [6, 7] = −1; [7, 8] = +1.
Net change $\int_0^8 f'(t)\,dt = 4 + 1 - 1 - 4 - 1 + 1 = 0$.
Therefore, f(8) = f(0) + 0 = **0**.

(d) f(3) = 0 + 4 + 1 = 5. f(7) = 5 − 1 − 4 − 1 = −1.
Comparing candidate values f(0) = 0, f(3) = 5, f(7) = −1, and f(8) = 0:
The **absolute maximum value is 5 at t = 3**; the **absolute minimum value is −1 at t = 7**.

AP SCORING CRITERIA

Evaluation Criterion	Points
Correct increasing intervals (0, 3) and (7, 8) and decreasing interval (3, 7)	2 pts
Correct relative maximum at t = 3 and relative minimum at t = 7 with sign justification	2 pts
Correct geometric signed-area setup and evaluation of f(8) = 0	3 pts
Correct absolute maximum 5 at t = 3 and absolute minimum −1 at t = 7	2 pts
Total	9 pts

Question 92: Mean Value Theorem

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) f is a polynomial, so it is continuous on the closed interval [0, 4] and differentiable on the open interval (0, 4). These satisfy the hypotheses of the Mean Value Theorem.

(b) $f(0) = 1$ and $f(4) = 16 - 16 + 1 = 1$.
Average rate of change $= [f(4) - f(0)] / (4 - 0) = (1 - 1) / 4 = \mathbf{0}$.

(c) $f'(t) = 2t - 4$. Set $f'(c) = 0$:
$2c - 4 = 0 \Rightarrow \mathbf{c = 2}$, which lies strictly inside (0, 4).

(d) The secant line slope connecting (0, 1) and (4, 1) is 0, and the tangent line slope at $t = 2$ is $f'(2) = 0$. Since both lines have equal slopes of 0, they are parallel (both are horizontal).

AP SCORING CRITERIA

Evaluation Criterion	Points
States continuity on [0, 4] and differentiability on (0, 4)	2 pts
Correct evaluation of average rate of change = 0	2 pts
Correct derivative equation $2c - 4 = 0$ and solution $c = 2$ in (0, 4)	3 pts
Explains that equal slopes (both 0) imply parallel lines	2 pts
Total	9 pts

Question 93: Absolute Extrema and the Candidates Test

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $g'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$.
Setting $g'(x) = 0$ gives the interior critical numbers **$x = 1$ and $x = 3$**.

(b) Evaluating candidate points:
- $g(0) = 0$
- $g(1) = 1 - 6 + 9 = 4$
- $g(3) = 27 - 54 + 27 = 0$
- $g(5) = 125 - 150 + 45 = 20$

(c) Comparing the candidate values {0, 4, 0, 20}:
The **absolute maximum value is 20 at $x = 5$**.
The **absolute minimum value is 0 at $x = 0$ and $x = 3$**.

AP SCORING CRITERIA

Evaluation Criterion	Points
Correct derivative and critical numbers $x = 1$ and $x = 3$	2 pts
Correct evaluation of all four candidate values: $g(0)=0$, $g(1)=4$, $g(3)=0$, $g(5)=20$	3 pts
Correct absolute maximum value 20 at $x = 5$	2 pts
Correct absolute minimum value 0 at $x = 0$ and $x = 3$	2 pts
Total	9 pts

Question 94: Concavity and Points of Inflection

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $f'(x) = x^3 - 2x^2$.
$f''(x) = 3x^2 - 4x =$ **$x(3x - 4)$**.

(b) The zeros of f'' are $x = 0$ and $x = 4/3$.
Sign of $f''(x)$: positive on $(-\infty, 0)$, negative on $(0, 4/3)$, and positive on $(4/3, \infty)$.
Therefore, f is **concave up on $(-\infty, 0) \cup (4/3, \infty)$** and **concave down on $(0, 4/3)$**.

(c) Concavity changes at $x = 0$ and $x = 4/3$.
$f(0) = 0$.
$f(4/3) = (1/4)(256/81) - (2/3)(64/27) = 64/81 - 128/81 = -64/81$.
The points of inflection are **$(0, 0)$ and $(4/3, -64/81)$**.

(d) A point of inflection requires an actual change in concavity across the point. For example, for $g(x) = x^4$, $g''(0) = 0$, but $g''(x) = 12x^2 \geq 0$ on both sides, so there is no change in concavity and no inflection point.

AP SCORING CRITERIA

Evaluation Criterion	Points
Correct first and second derivatives and factored form $x(3x - 4)$	2 pts
Correct concavity intervals with sign justification	2 pts
Correct coordinates $(0, 0)$ and $(4/3, -64/81)$	3 pts
Explains requirement that f'' must change sign / concavity must change	2 pts
Total	9 pts

Question 95: Second Derivative Test

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $f'(x) = 4x^3 - 8x = 4x(x^2 - 2)$.
Setting $f'(x) = 0$ yields critical numbers **$x = 0$, $x = -\sqrt{2}$, and $x = \sqrt{2}$.**
$f''(x) = 12x^2 - 8$.

(b) Applying the Second Derivative Test:
- At $x = 0$: $f''(0) = -8 < 0 \Rightarrow$ **relative maximum**.
- At $x = -\sqrt{2}$: $f''(-\sqrt{2}) = 12(2) - 8 = 16 > 0 \Rightarrow$ **relative minimum**.
- At $x = \sqrt{2}$: $f''(\sqrt{2}) = 12(2) - 8 = 16 > 0 \Rightarrow$ **relative minimum**.

(c) Function values:
- $f(0) = 0 \Rightarrow$ **relative maximum at (0, 0)**.
- $f(-\sqrt{2}) = 4 - 8 = -4 \Rightarrow$ **relative minimum at $(-\sqrt{2}, -4)$**.
- $f(\sqrt{2}) = 4 - 8 = -4 \Rightarrow$ **relative minimum at $(\sqrt{2}, -4)$**.

AP SCORING CRITERIA

Evaluation Criterion	Points
Correct derivatives f', f'' and critical numbers $0, \pm\sqrt{2}$	**3 pts**
Correct evaluation of f'' at each critical point with correct classifications	**3 pts**
Correct coordinate points: max $(0, 0)$, mins $(\pm\sqrt{2}, -4)$	**3 pts**
Total	**9 pts**

Question 96: Using a Sign Table for f′

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) f is **increasing on (−2, −1) and (1, 3)** because f′ > 0 there.
f is **decreasing on (−1, 1)** because f′ < 0 there.

(b) In the interior (−2, 3):
- At x = −1, f′ changes from + to −, so f has a **relative maximum of 4 at x = −1**.
- At x = 1, f′ changes from − to +, so f has a **relative minimum of −2 at x = 1**.

(c) Using the Candidates Test on [−2, 3]:
Candidate values are f(−2) = 0, f(−1) = 4, f(1) = −2, and f(3) = 1.
The **absolute maximum value is 4 at x = −1**.
The **absolute minimum value is −2 at x = 1**.

AP SCORING CRITERIA

Evaluation Criterion	Points
Correct intervals of increase and decrease with justification	3 pts
Relative maximum at x = −1 and relative minimum at x = 1 with First Derivative Test justification	2 pts
Candidate comparison and absolute maximum 4 at x = −1	2 pts
Candidate comparison and absolute minimum −2 at x = 1	2 pts
Total	9 pts

Question 97: Optimization: Fencing Beside a River

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) The total fencing is $2x + y = 80 \Rightarrow y = 80 - 2x$.
Area is $A(x) = xy = \mathbf{x(80 - 2x) = 80x - 2x^2}$.
Physical domain: $x > 0$ and $80 - 2x > 0 \Rightarrow \mathbf{0 < x < 40}$.

(b) $A'(x) = 80 - 4x$. Setting $A'(x) = 0$ gives:
$80 - 4x = 0 \Rightarrow \mathbf{x = 20}$ **meters**.
Then $y = 80 - 2(20) =$ **40 meters**.

(c) $A''(x) = -4 < 0$ for all x, so by the Second Derivative Test, A has an absolute maximum at $x = 20$.
Maximum area $= A(20) = 20(40) =$ **800 square meters**.

AP SCORING CRITERIA

Evaluation Criterion	Points
Correct constraint $2x + y = 80$, area function $A(x) = 80x - 2x^2$, and domain $(0, 40)$	2 pts
Differentiates to find $A' = 80 - 4x$ and solves $x = 20$, $y = 40$	4 pts
Justifies maximum ($A'' < 0$) and states maximum area 800 m²	3 pts
Total	9 pts

Question 98: Optimization: Open-Top Storage Box

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) Height = x, base dimensions = 12 − 2x by 12 − 2x.
Volume: $\mathbf{V(x) = x(12 - 2x)^2 = 4x(6 - x)^2}$.
Physical domain: $\mathbf{0 < x < 6}$.

(b) Expanding: $V(x) = x(144 - 48x + 4x^2) = 4x^3 - 48x^2 + 144x$.
$V'(x) = 12x^2 - 96x + 144 = 12(x^2 - 8x + 12) = 12(x - 2)(x - 6)$.
Setting $V'(x) = 0$ gives $x = 2$ and $x = 6$. The only critical point in $(0, 6)$ is **$x = 2$ inches**.

(c) $V'(x)$ is positive for $0 < x < 2$ and negative for $2 < x < 6$, so by the First Derivative Test, V has a maximum at $x = 2$.
Box dimensions: length = 8 in, width = 8 in, height = 2 in.
Maximum volume = $V(2) = 2(8)(8) =$ **128 cubic inches**.

AP SCORING CRITERIA

Evaluation Criterion	Points
Correct volume expression $V(x) = x(12 - 2x)^2$ and domain $(0, 6)$	2 pts
Correct derivative $V' = 12(x - 2)(x - 6)$ and critical point $x = 2$	3 pts
Justifies maximum (sign change of V') and states dimensions 8×8×2 in and volume 128 in^3	4 pts
Total	9 pts

Question 99: Behavior of an Implicit Curve

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) Differentiating implicitly: $2x + 2y(dy/dx) = 0 \Rightarrow$ **$dy/dx = -x / y$** (for $y \neq 0$).

(b) On the upper semicircle, $y > 0$ for x in $(-5, 5)$.
• For $-5 < x < 0$: $dy/dx = -(-)/+ > 0$, so the curve is **increasing on $(-5, 0)$**.
• For $0 < x < 5$: $dy/dx = -(+)/+ < 0$, so the curve is **decreasing on $(0, 5)$**.

(c) The **absolute maximum value is $y = 5$ at $(0, 5)$**.
The **absolute minimum value is $y = 0$ at $(-5, 0)$ and $(5, 0)$**.

(d) Vertical tangents occur where dy/dx is undefined with a nonzero numerator, which occurs when $y = 0$. On $x^2 + y^2 = 25$, $y = 0$ yields $x = \pm 5$. Thus, vertical tangents occur at **$(-5, 0)$ and $(5, 0)$**.

AP SCORING CRITERIA

Evaluation Criterion	Points
Correct implicit differentiation yielding $dy/dx = -x/y$	2 pts
Correct sign analysis and intervals: increasing on $(-5, 0)$, decreasing on $(0, 5)$	3 pts
Correct absolute maximum $y = 5$ at $(0, 5)$ and absolute minimum $y = 0$ at $(\pm 5, 0)$	2 pts
Identifies points $(-5, 0)$ and $(5, 0)$ where denominator vanishes and states vertical tangent	2 pts
Total	9 pts

Question 100: Optimization: Inscribed Rectangle Under a Parabola

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) The width of the rectangle is $2x$ and its height is $12 - x^2$.
Area: $\mathbf{A(x) = 2x(12 - x^2) = 24x - 2x^3}$.
Domain: $\mathbf{0 \le x \le \sqrt{12}}$ (or $0 \le x \le 2\sqrt{3}$).

(b) Differentiating: $A'(x) = 24 - 6x^2$.
Setting $A'(x) = 0$: $6x^2 = 24 \Rightarrow x^2 = 4 \Rightarrow \mathbf{x = 2}$ (since $x \ge 0$).
At $x = 0$ and $x = \sqrt{12}$, $A(x) = 0$. Since $A'(x) > 0$ for $0 \le x < 2$ and $A'(x) < 0$ for $x > 2$, $x = 2$ yields the absolute maximum.

(c) At $x = 2$:
- Width $= 2(2) =$ **4 units**.
- Height $= 12 - (2^2) = 12 - 4 =$ **8 units**.
- Maximum area $= A(2) = 4 \times 8 =$ **32 square units**.

AP SCORING CRITERIA

Evaluation Criterion	Points
Correct area function $A(x) = 2x(12 - x^2)$ and domain $[0, \sqrt{12}]$	2 pts
Differentiates to find $A' = 24 - 6x^2$ and solves $x = 2$ with maximum justification	4 pts
Correct dimensions: width = 4, height = 8, and maximum area = 32	3 pts
Total	9 pts

Essential Unit 5 Formulas and Theorems

- **Mean Value Theorem (MVT):** If f is continuous on [a, b] and differentiable on (a, b), there exists at least one c in (a, b) such that $f'(c) = [f(b) - f(a)] / (b - a)$.
- **Rolle's Theorem:** If in addition $f(a) = f(b)$, then $f'(c) = 0$ for some c in (a, b).
- **Extreme Value Theorem (EVT):** A continuous function on a closed bounded interval [a, b] attains both an absolute maximum and an absolute minimum on [a, b].
- **Critical Numbers:** Values c in the domain of f where $f'(c) = 0$ or $f'(c)$ does not exist.
- **First Derivative Test:** If f' changes from + to − at c, f has a local maximum; if f' changes from − to +, f has a local minimum. If f' does not change sign, there is no local extremum.
- **Candidates Test for Absolute Extrema:** Evaluate f at all domain endpoints and all interior critical numbers. The largest value is the absolute maximum; the smallest is the absolute minimum.
- **Concavity & Points of Inflection:** $f''(x) > 0$ implies concave up; $f''(x) < 0$ implies concave down. An inflection point requires an actual sign change of $f''(x)$.
- **Second Derivative Test:** If $f'(c) = 0$ and $f''(c) > 0$, f has a local minimum at c. If $f'(c) = 0$ and $f''(c) < 0$, f has a local maximum. If $f''(c) = 0$, the test is inconclusive.
- **Optimization Strategy:** 1) Identify objective function and constraint; 2) Express objective in terms of a single variable; 3) State feasible domain; 4) Find critical points and justify global extremum using derivative tests or endpoints.

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